# BIM

## PORTFOLIO

VLADIMIR GARKUSHENKO

BIM MANAGER I ML ENGINEER

2026

# ABOUT

## VLADIMIR GARKUSHENKO

BIM Manager I ML Engineer



I focus on multidisciplinary coordination, BIM standards, and structured Revit-based project delivery. My experience includes developing BIM execution frameworks, setting up project environments, managing model coordination, creating families and templates, and maintaining quality and consistency across design teams. In addition to core BIM management, I apply automation and data-driven methods through Dynamo, Revit API workflows, metadata extraction, and ML-oriented BIM data analysis to support more efficient, reliable, and scalable project delivery.
I combine BIM management, coordination, automation, and data analysis to support scalable project delivery and BIM standardization.

## Work Experience

**BIM ML Engineer**
SVOD — 2026 - Present

**BIM Manager**
BAUMER — 2025 - 2026

**BIM Manager**
RPA HEIGHT — 2022 - 2025

**BIM Coordinator**
ADAMANT — 2021 - 2022

## Contact

garkushenkovladimir@gmail.com

www.linkedin.com/in/vladimir-garkushenko

## Education

**Peter the Great St. Petersburg Polytechnic University**
Bachelor of Science in Civil Engineering
2015 – 2019

## Skills

BIM & Standards: ISO 19650, BEP, EIR, Information Management, BIM Governance, CDE workflows

Automation: Dynamo, Revit API (C#), Python (NumPy, Pandas, PyTorch), GitHub, Visual Studio Code

Software: Revit, Navisworks, AutoCAD, MagiCAD, Blender + Bonsai (IFC), FARO, Power BI

## Language Proficiency

English - Upper-Intermediate (B2) I Russian – Native I French - Beginner

# CONTENTS


BIM Technology Stack 01

Revit Projects 02

FARO Laser Scanning 10

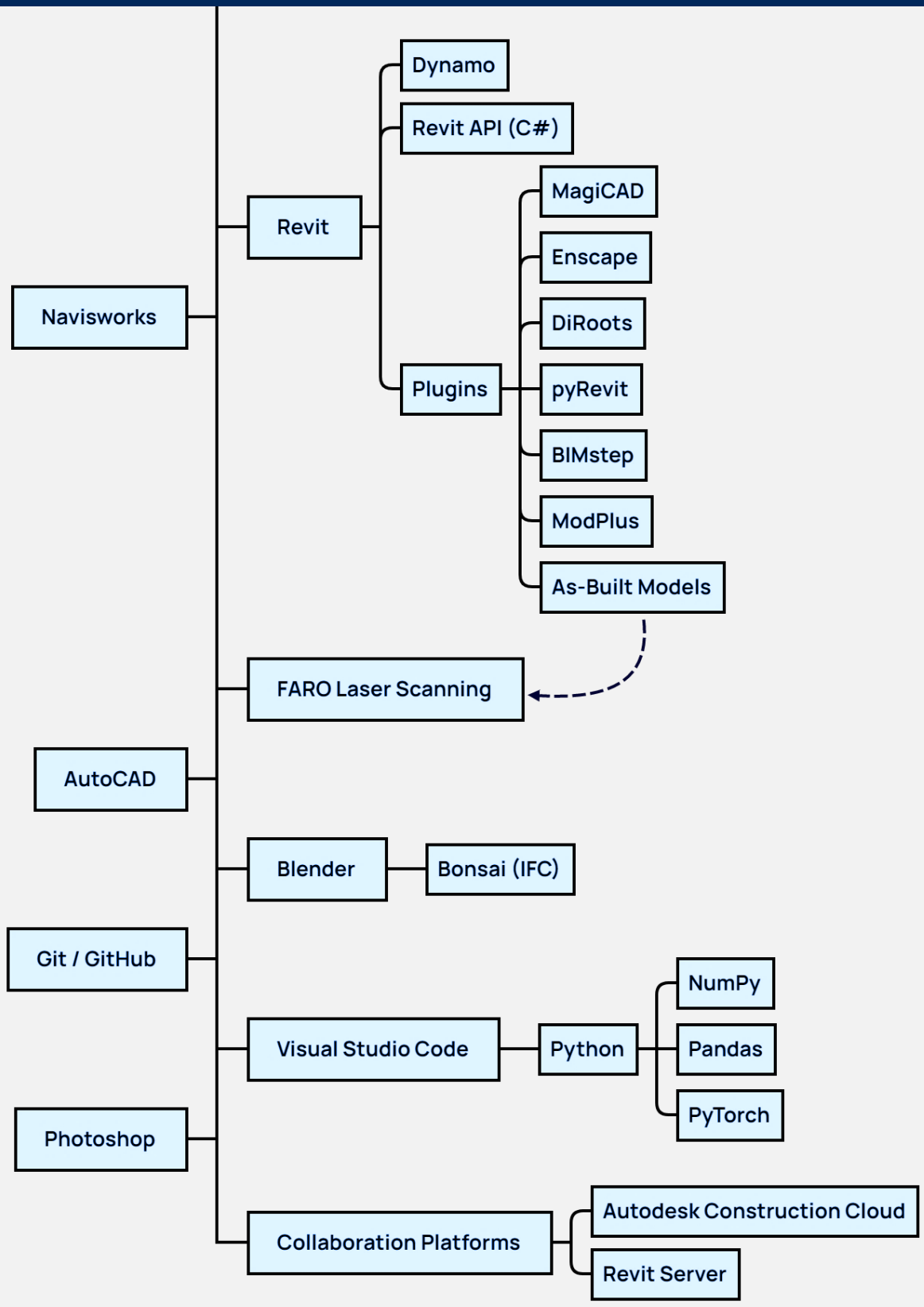
Revit Families 11

Navisworks 13

Dynamo Scripts 14

ML-Based Data Analysis System 15

# BIM Technology Stack

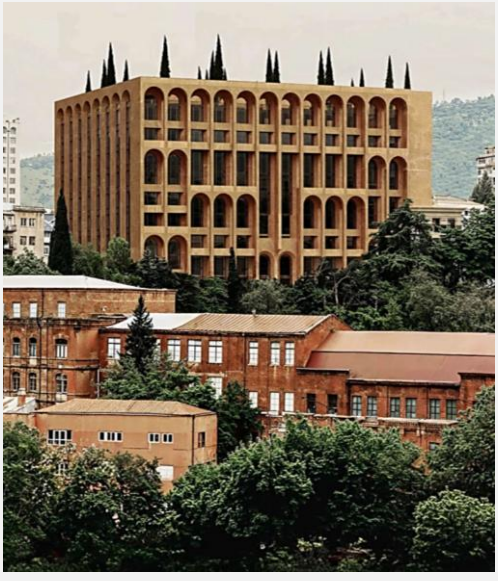


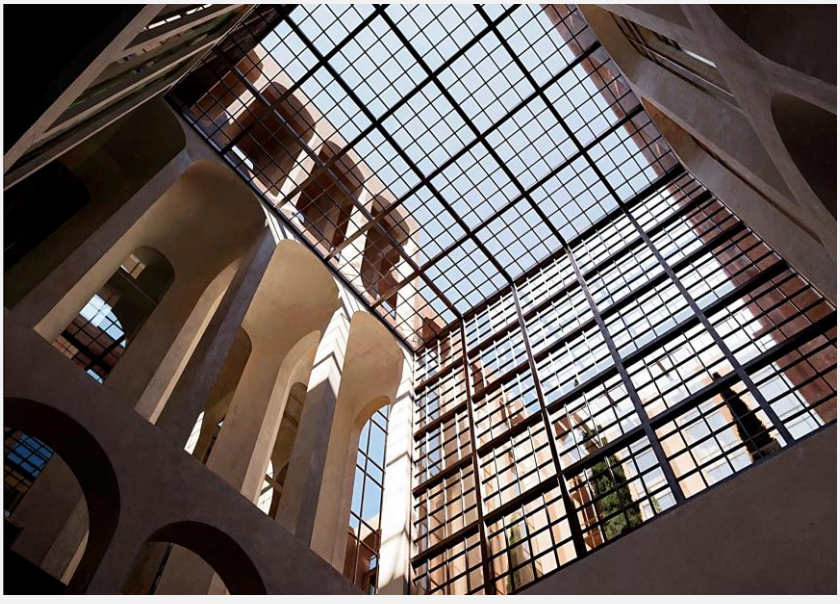
# Apartment Hotel Kostava

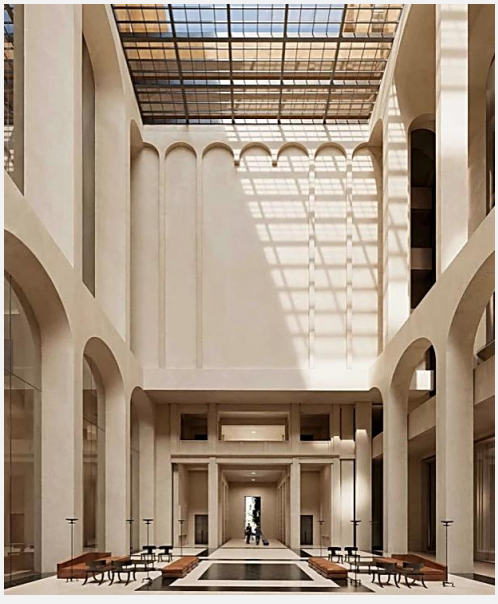
Role: BIM Manager  Software: Revit 2024  Disciplines: Architecture, Structure, MEP

High-end apartment hotel in Tbilisi, delivered as a fully coordinated multidisciplinary project within a BIM environment. The project required linked model integration, cross-disciplinary coordination, and consistent design management to ensure quality, consistency, and technical accuracy throughout the design process.







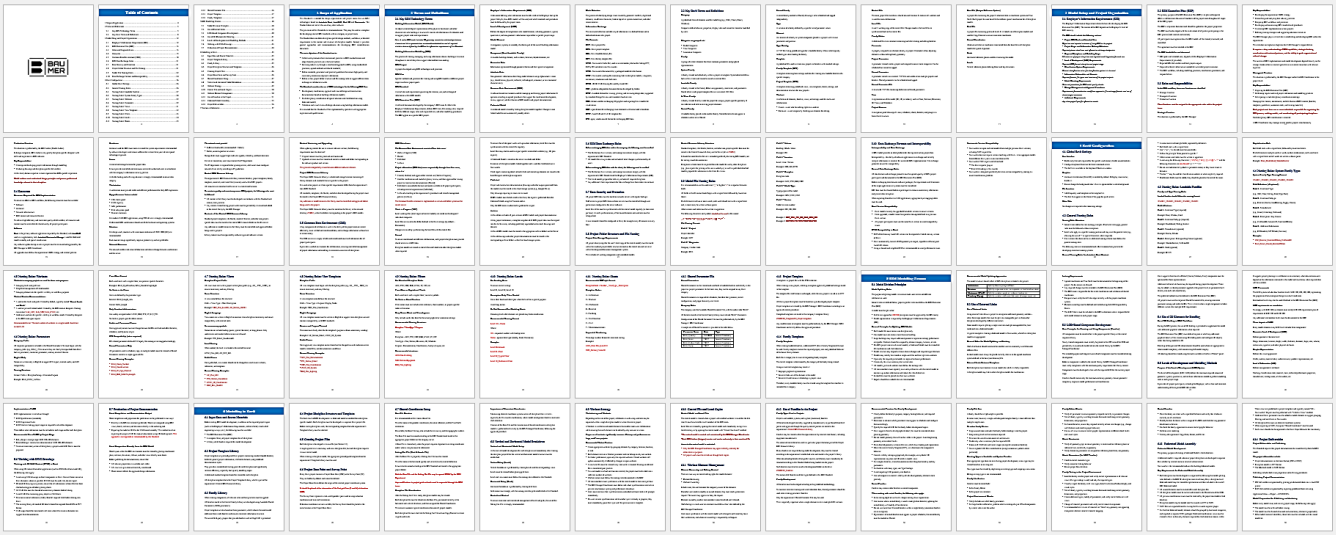


# Apartment Hotel Kostava

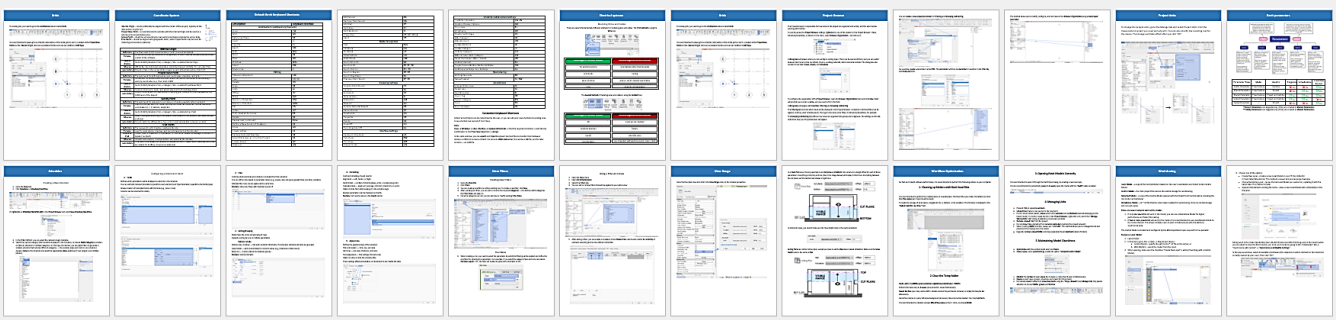
Established BIM execution workflows and structured Revit environments for a 21,000 m² multidisciplinary project comprising 14 models and involving around 50 specialists, supporting predictable collaboration and reducing coordination issues across disciplines. Delivered BIM training sessions for project teams, helping accelerate onboarding and improve workflow consistency. Created and maintained Revit families to support consistent data structure, coordinated documentation sets, and schedule generation.

Developed a BIM standard and role-specific guidance package totaling over 100 pages, aligned with international requirements and used across all specialist teams, covering model structure, naming rules, worksets, shared coordinates, views, deliverables, and quality control procedures.

BIM Standard



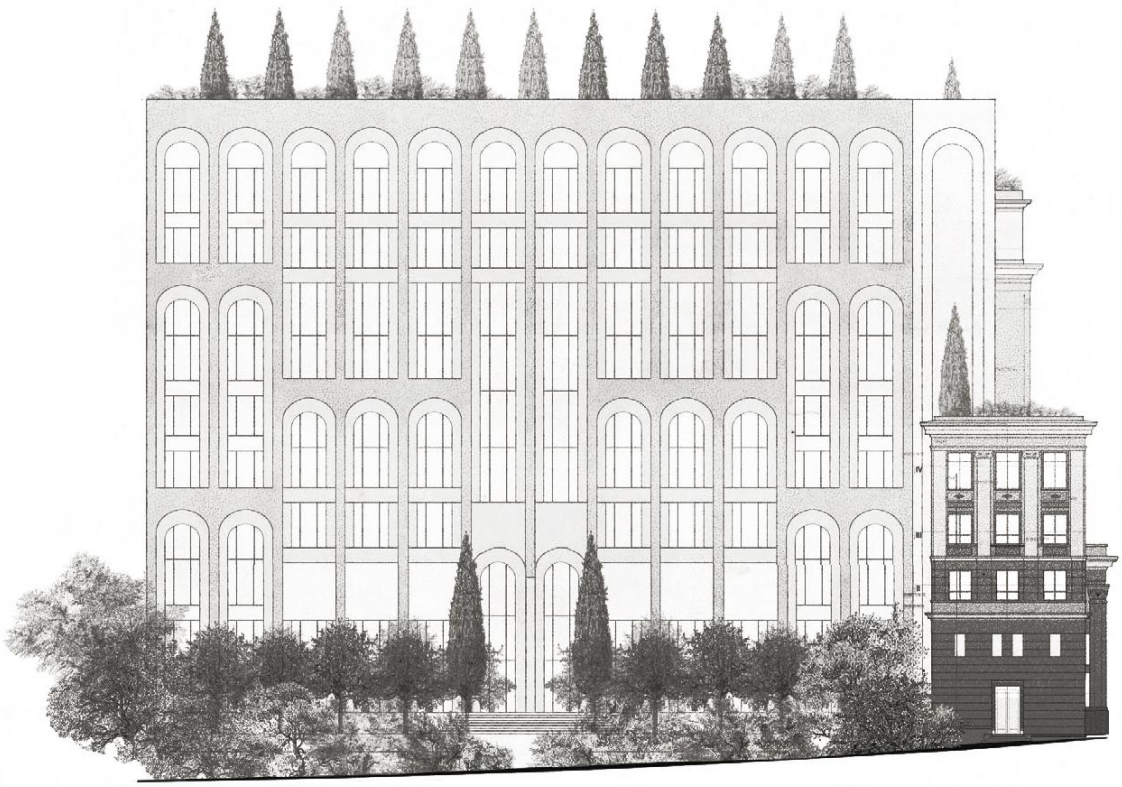
Instructions



# Apartment Hotel Kostava

Architectural concept sketch





# Apartment Hotel Kostava

Rendered views





# Dau House Residential Tower

Role: BIM Manager I Software: Revit 2022 I Disciplines: Architecture

Premium residential tower in Moscow. Rising 340 meters, it is one of the tallest residential buildings. The project scope included architectural model development and architectural documentation. The development includes a private landscaped site, its own park, and underground parking.









# Dau House Residential Tower

Built and maintained a unified BIM environment for 12 Revit-based architectural projects with a total area of 105,000 m², improving data consistency and coordination across project teams.
Standardized shared parameters, families, and model structure to ensure consistency across project environments.
Managed weekly clash detection and reporting in Navisworks, supporting continuous coordination and tracking a high-volume process involving over 100,000 clash records.
Applied Dynamo scripts and Revit plugins to streamline routine checks, reduce manual effort, and improve team efficiency.

Rendered views







# Hosser Rehabilitation Center

Role: BIM Manager  Software: Revit 2022  Disciplines: Architecture, Structure, MEP

The Hosser Rehabilitation Center in Moscow was delivered as a coordinated multidisciplinary project within a structured BIM environment. The project was organized into five design sections and required precise model coordination, shared project standards, and quality control procedures to ensure consistency and compliance throughout the design process.



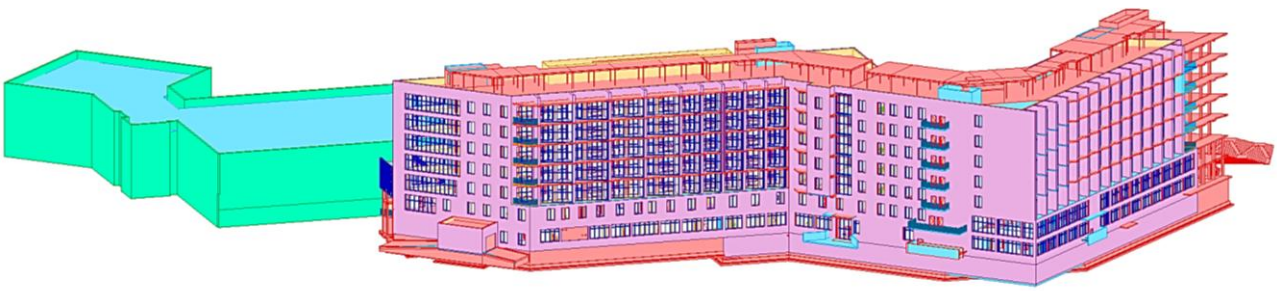




# Hosser Rehabilitation Center

For a project of over 45,000 m² with 15 Revit models, I translated client requirements into a structured BIM execution framework and configured Revit environments to ensure consistent project setup, data structure, and team workflows.
I established coordination rules for linked models and carried out weekly clash detection in Navisworks, contributing to the resolution of over 20,000 clashes and improving issue management across disciplines. I also implemented Dynamo-based checks and refined clash test logic, excluding more than 50,000 false-positive clash results, increasing data completeness, and reducing manual quality control effort.

Visualization of Project Worksets Architecture and Structure



Visualization of Project Worksets MEP



# Cultural Heritage Museum

Role: BIM Manager  Software: Revit 2022  Disciplines: Architecture, Structure, MEP

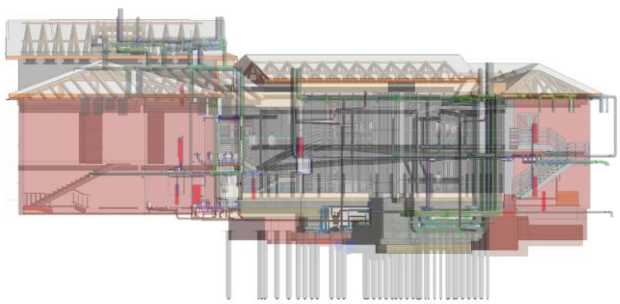
Project for the preservation and adaptive reuse of a cultural heritage building in Tula. Using FARO 3D laser scanning data and the As-Built plug-in, I supervised point cloud workflows and supported development of an accurate as-built Revit model for reconstruction and repair works. I created Revit models, templates, schedules, filters, and worksets, enabling structured documentation, material quantity extraction, and consistent information quality across the project.

Laser scanning point cloud



Revit project

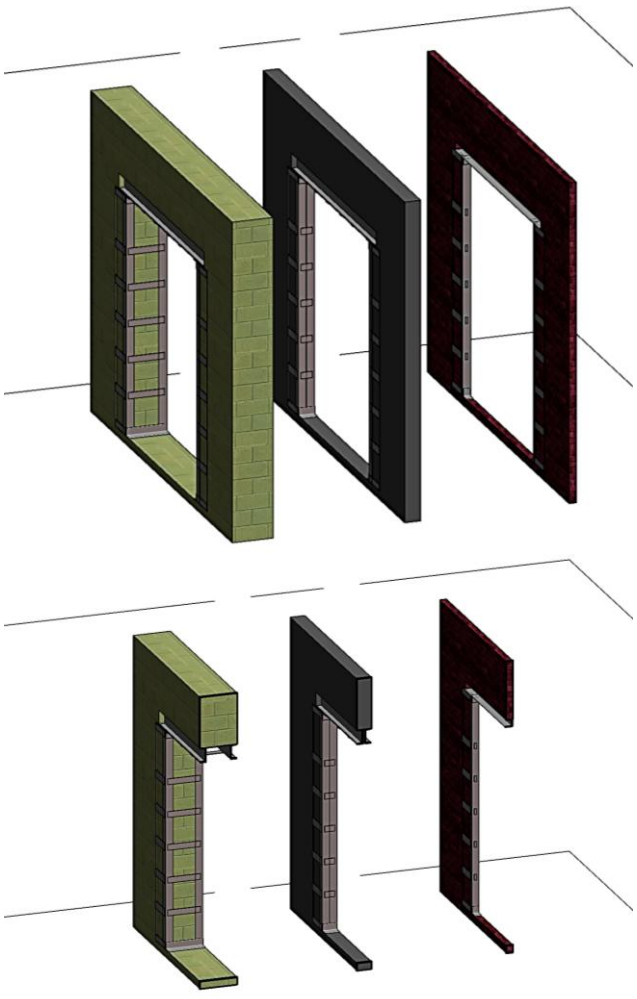




# Revit Families

Developed several Revit family libraries comprising hundreds of families across the architectural, structural, and MEP disciplines, as well as annotation families. These families were created to the required LODs and included all necessary parameters, family types, and lookup tables. Additionally, I developed parametric Revit families that automatically generate material schedules and Bills of Quantities through nested families and embedded formulas. One example is an opening reinforcement family that adapts to wall thickness and opening dimensions, automatically placing plates, angles, channels, and related components while populating the required schedule data.

Structural opening reinforcement family



Nested families




<Material schedule>

| A | B | C | D |
|---|---|---|---|
| Product name | Quantity, pcs. | Length of products, m | Volume |
| | | | |
| Angle 100x100x8 8509-93 | 63 | 42.1 | 0.066 m³ |
| Angle 100x100x8 8509_93 | 54 | 15.1 | 0.024 m³ |
| Channel 10P 8240-97 | 18 | 54.0 | 0.059 m³ |
| Channel 16P 8240-97 | 36 | 108.0 | 0.407 m³ |
| Plate 5x80 19903-2015 | 270 | 59.4 | 0.024 m³ |
| Plate 5x80 19903-2015 - Horizontal | 72 | 27.0 | 0.011 m³ |
| Plate 5x80x360 19903-2015 | 216 | 77.8 | 0.031 m³ |
| Plate 5x100x360 19903-2015 | 540 | 194.4 | 0.097 m³ |

# Revit Families

A key part of my Revit family development approach is configuring nested families and embedding logic aligned with real design and construction workflows, including the use of family types and lookup tables. This allows material quantities and work scopes to be calculated automatically at placement stage, improving schedule accuracy and significantly reducing preparation time.

## Family parameters

| Parameter | Value | Formula |
|---|---|---|
| **Constraints** | | |
| Default Elevation | 0.0 | = |
| **Dimensions** | | |
| T (default) | 100.0 | = Wall thickness |
| t2 (default) | 50.0 | = Wall thickness / 2 |
| Height (default) | 3000.0 | = |
| Height of the upper notch (default) | 100.0 | = if(T > 120 mm, 200 mm, 100 mm) |
| Length of inner plate (default) | 40.0 | = t2 * 2 - 60 mm |
| Channel length (default) | 3500.0 | = Width + 500 mm |
| Frame clearance | 10.0 | = |
| Floor mark | 100.0 | = |
| Plate offset | 150.0 | = |
| Distance between plates | 450.0 | = |
| Wall thickness (report) | 100.0 | = |
| Width (default) | 3000.0 | = |
| **Visibility** | | |
| Steel jacket (default) | ☑ | = |
| Right reinforcement (default) | ☑ | = |
| Left reinforcement (default) | ☑ | = |
| **Overall Legend** | | |
| Lintel installation (default) | Lintel installation (L100x8 angle) | = if(T > 120 mm, "Lintel installation (Channel No.16)", "Lintel installation (L100x8 angle)") |
| Lintel installation length (default) | 3500.0 | = if(T > 120 mm, Width + 500 mm, Width + 500 mm) |
| Quantity of lintel installation (default) | 1.000000 | = if(T > 120 mm, 2, 1) |
| Hilti HIT HY 70 Ø16 studs L=200 mm for steel jacket fixing with chemical ancho | | = if(T > 120 mm, "Ø16 studs for steel jacket fixing with chemical anchors Hilti HIT HY 70", "") |
| Hilti HIT HY 70 Ø16 studs L=200 mm for steel jacket fixing (length) (default) | 200.0 | = 200 mm |
| Quantity of Ø16 studs L=200 mm for steel jacket fixing (default) | 0.000000 | = if(T > 120 mm, if(round(Height / 1000 mm / 0.5 * 10) - round(Height / 1000 mm / 0.5) * 10 < 3, 2 * round(Height / 1000 mm / 0.5 - ((Height / 1000 mm / 0.5 * 10) - round(Height / 1000 mm / 0.5) * 10) / 10), 2 * roundup(Height / 1000 mm / 0.5)), 0) |
| Ø16 studs L=250 mm for steel jacket fixing (default) | | = if(T = 510 mm, "Ø16 studs for steel jacket fixing", "") |
| Ø16 studs L=250 mm for steel jacket fixing (length) (default) | 0.0 | = if(T = 510 mm, 250 mm, 0 mm) |
| Quantity of Ø16 studs L=250 mm for steel jacket fixing (default) | 0.000000 | = if(T = 510 mm, if(round(Height / 1000 mm / 0.5 * 10) - round(Height / 1000 mm / 0.5) * 10 < 3, 4 * round(Height / 1000 mm / 0.5 - ((Height / 1000 mm / 0.5 * 10) - round(Height / 1000 mm / 0.5) * 10) / 10), 4 * roundup(Height / 1000 mm / 0.5)), 0) |
| Ø16 studs L=100 mm for steel jacket fixing (default) | | = if(T < 510 mm, if(T > 249 mm, "Ø16 studs for steel jacket fixing", ""), "") |
| Ø16 studs L=100 mm for steel jacket fixing (length) (default) | 0.0 | = if(T < 510 mm, if(T > 249 mm, T + 100 mm, 0 mm), 0 mm) |
| Quantity of Ø16 studs L=100 mm for steel jacket fixing (default) | 0.000000 | = if(T < 510 mm, if(T > 249 mm, if(round(Height / 1000 mm / 0.5 * 10) - round(Height / 1000 mm / 0.5) * 10 < 3, 2 * round(Height / 1000 mm / 0.5 - ((Height / 1000 mm / 0.5 * 10) - round(Height / 1000 mm / 0.5) * 10) / 10), 2 * roundup(Height / 100 |
| Wall chasing in brickwork (default) | Brickwork chasing 20 mm deep for installation | = if(T > 120 mm, "Brickwork chasing 70 mm deep and 170 mm wide for installation of Channel No.16, length", "Brickwork chasing 20 mm deep for installation of L100x8 angle") |
| Brickwork chasing length (default) | 3500.0 | = if(T > 120 mm, Width + 500 mm, Width + 500 mm) |
| Quantity of brickwork chasing (default) | 0.018000 | = if(T > 120 mm, (0.024 * Width / 1 mm + 0 mm * 500 / 1 mm) / 1000, (0.006 * Width / 1 mm + 0 mm * 500 / 1 mm) / 1000) |
| Brick wall demolition (default) | Brick wall demolition | = "Brick wall demolition" |
| Brick wall demolition length (default) | 3000.0 | = if(T > 120 mm, Width, Width) |
| Quantity of brick wall demolition (default) | 0.900000 | = Width * Height * T / 1 m³ |
| Ø12 studs L=200 mm for steel jacket fixing (default) | Ø12 studs for steel jacket fixing | = if(T < 121 mm, "Ø12 studs for steel jacket fixing", "") |
| Ø12 studs L=200 mm for steel jacket fixing (length) (default) | 200.0 | = if(T < 121 mm, 200 mm, 0 mm) |
| Quantity of Ø12 studs L=200 mm for steel jacket fixing (default) | 12.000000 | = if(T < 121 mm, if(round(Height / 1000 mm / 0.5 * 10) - round(Height / 1000 mm / 0.5) * 10 < 3, 2 * round(Height / 1000 mm / 0.5 - ((Height / 1000 mm / 0.5 * 10) - round(Height / 1000 mm / 0.5) * 10) / 10), 2 * roundup(Height / 1000 mm / 0.5)), 0) |
| Ø16 studs L=100 mm, spacing 500 mm (default) | | = if(T > 120 mm, "Ø16 studs at 500 mm spacing", "") |
| Ø16 studs L=100 mm, spacing 500 mm (length) (default) | 0.0 | = if(T > 120 mm, T - 100 mm, 0 mm) |
| Quantity of Ø16 studs L=100 mm, spacing 500 mm (default) | 0.000000 | = if(T > 120 mm, round((Width / 1000 mm + 0.5) / 0.4), 0) |
| Drilling Ø18 holes in brickwork (depth t mm) (default) | | = if(T > 120 mm, "Drilling Ø18 holes in brickwork, depth", "") |
| Drilling Ø18 holes in brickwork (length) (default) | 0.0 | = if(T > 120 mm, T, 0 mm) |
| Quantity of drilled Ø18 holes in brickwork (default) | 0.000000 | = if(T > 120 mm, round((Width / 1000 mm + 0.5) / 0.4), 0) |
| Anti-corrosion coating of steel structures (default) | Anti-corrosion coating of steel structures | = "Anti-corrosion coating of steel structures" |
| Anti-corrosion coating length (default) | 0.0 | = |
| Quantity of anti-corrosion coating (default) | 0.000000 | = |
| Grouting of lintel with cement-sand mortar (default) | Lintel grouting with cement-sand mortar | = "Lintel grouting with cement-sand mortar" |
| Lintel grouting length (default) | 0.0 | = |
| Quantity of lintel grouting (default) | 0.011000 | = if(T < 120.1 mm, round(0.003 * (Width + 500 mm) / 1 mm) / 1000, round(0.016 * (Width + 500 mm) / 1 mm) / 1000) |
| Plastering of lintel and steel jacket over mesh 50x50 (default) | Plastering of lintel and steel jacket over Ø3 me | = "Plastering of lintel and steel jacket over Ø3 mesh, 50x50 spacing" |
| Plastering length (lintel and steel jacket over mesh 50x50) (default) | 0.0 | = |
| Quantity of plastering (lintel and steel jacket over mesh 50x50) (default) | 2.630000 | = if(T < 121 mm, round((0.94 * Height / 1000 mm * Width / 1000 mm * 0.24 + 600 mm / 1000 mm) * 100) / 100, (0.34 + T / 1000 mm) * Width / 1000 mm + 0.17 + (0.82 + T / 1000 mm) * Height / 1000 mm * 2) |
| **Other** | | |
| Plate quantity (default) | 5 | = (Height / 1 mm - 100) / 450 - 1 |
| Plate quantity under channel (default) | 3 | = (Width / 1 mm - 600) / 400 / 2 |
| Top plate (default) | 0.000000 | = if(Top steel jacket, 1, 0) |
| Left plate (default) | 1.000000 | = if(Left steel jacket, 1, 0) |
| Right plate (default) | 1.000000 | = if(Right steel jacket, 1, 0) |
| Channel 10P (default) | 1.000000 | = if(T < 121 mm, 1, 0) |
| Channel 16P (default) | 0.000000 | = if(T > 120 mm, 1, 0) |
| Top steel jacket (default) | ☐ | = and(T > 120 mm, Steel jacket) |
| Left steel jacket (default) | ☑ | = and(Left reinforcement, Steel jacket) |
| Right steel jacket (default) | ☑ | = and(Right reinforcement, Steel jacket) |
| Left side channel (default) | ☑ | = and(T < 121 mm, Right reinforcement, Steel jacket) |
| Right side channel (default) | ☑ | = and(T < 121 mm, Left reinforcement, Steel jacket) |
| Top angle (default) | ☑ | = and(T < 121 mm, Steel jacket) |
| Side angles (default) | ☐ | = and(T > 120 mm, Steel jacket) |
| **Identity Data** | | |

# Navisworks

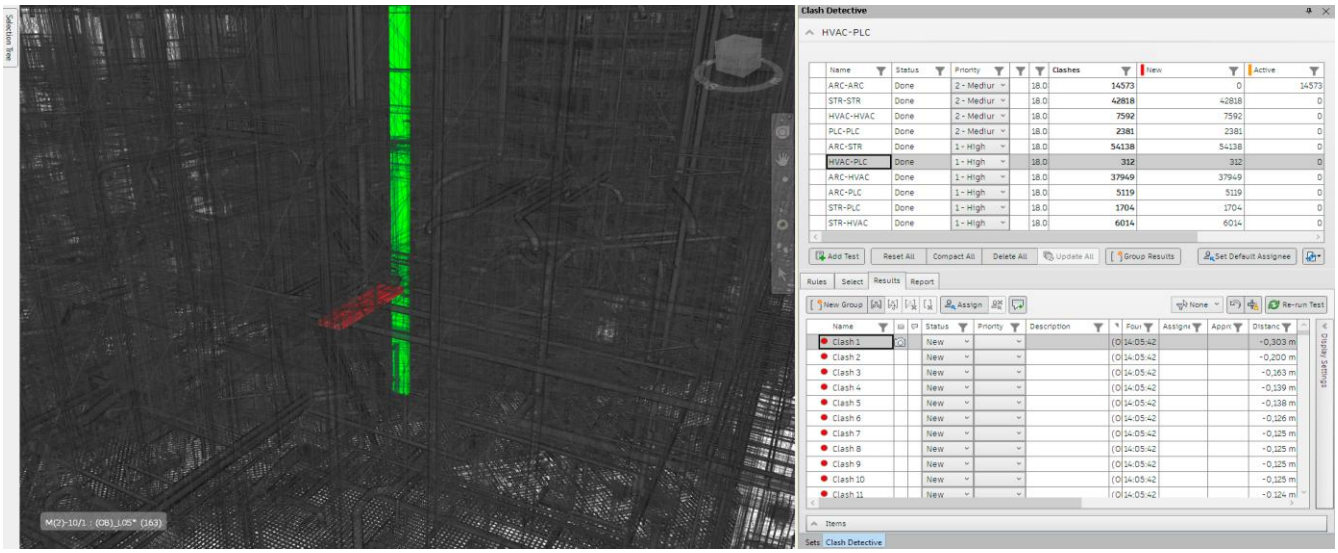
Performed multidisciplinary coordination and clash detection, improving issue identification and resolution workflows.

Configured clash tests, search sets, and exclusion rules, reducing false positives and increasing analysis accuracy.

Structured and tracked clashes, improving communication and coordination across disciplines.

Generated reports and monitored resolution status, increasing transparency and control.

Clash detection in Navisworks




Clash report

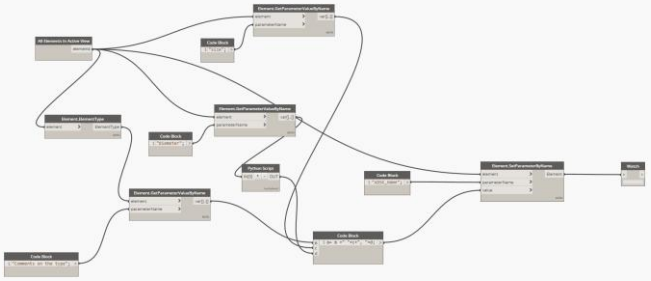
| HVAC-PLC | Tolerance | Clashes | New | Active | Reviewed | Approved | Resolved | Type | Status |
|---|---|---|---|---|---|---|---|---|---|
| | 0.005m | 312 | 312 | 0 | 0 | 0 | 0 | Hard | OK |

| | | | | | | Item 1 | | | | Item 2 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Image | Clash Name | Status | Distance | Grid Location | Clash Point | Item ID | Layer | Item Name | Item Type | Item ID | Layer | Item Name | Item Type |
|  | Clash 1 | New | -0.303 | L-13/1 : (HVAC)_L03* | x:2131586.542, y:454604.817, z:170.955 | Object ID: 5144745 | (HVAC)_L02 | Galvanized Steel | Solid | Object ID: 1857206 | L02 | Shaft_HVAC | Solid |
|  | Clash 2 | New | -0.200 | L-13/1 : (HVAC)_L02* | x:2131587.689, y:454605.011, z:167.230 | Object ID: 3763313 | (HVAC)_L01 | Shaft_Pressurization | Solid | Object ID: 1857425 | L02 | Shaft_HVAC | Solid |
|  | Clash 3 | New | -0.163 | D/1-9/1 : (HVAC)_L06 | x:2131600.025, y:454612.241, z:185.143 | Object ID: 6367464 | (HVAC)_L06 | CASING | Solid | Object ID: 1201461 | L06 | Porcelain | Solid |
|  | Clash 4 | New | -0.139 | L-13/1 : (HVAC)_L07 | x:2131587.757, y:454605.021, z:192.630 | Object ID: 3094740 | (HVAC)_L06 | ROCKWOOL_WIRED MAT 105 | Solid | Object ID: 1857425 | L02 | Shaft_HVAC | Solid |
|  | Clash 5 | New | -0.138 | 7/1"-H : (HVAC)_L03 | x:2131569.377, y:454609.129, z:171.780 | Object ID: 6830214 | (HVAC)_L01 | Shaft_HVAC | Solid | Object ID: 1211999 | L03 | Laminate - Linen,Matte | Solid |
|  | Clash 6 | New | -0.126 | G/1-9/1 : (HVAC)_L01* | x:2131599.602, y:454608.863, z:166.555 | Object ID: 6516223 | (HVAC)_L01 | Galvanized Steel | Solid | Object ID: 1832917 | L01 | Shaft_HVAC | Solid |
|  | Clash 7 | New | -0.125 | 7/1"-H : (HVAC)_L06 | x:2131569.293, y:454609.031, z:184.380 | Object ID: 6830214 | (HVAC)_L01 | Shaft_HVAC | Solid | Object ID: 1986940 | L06 | Laminate - Linen,Matte | Solid |
|  | Clash 8 | New | -0.125 | 10"-K/1" : (HVAC)_L03 | x:2131583.019, y:454635.168, z:172.085 | Object ID: 4475608 | (HVAC)_L01* | Shaft_HVAC | Solid | Object ID: 1626755 | L03* | RGB_RAL 9003 | Solid |

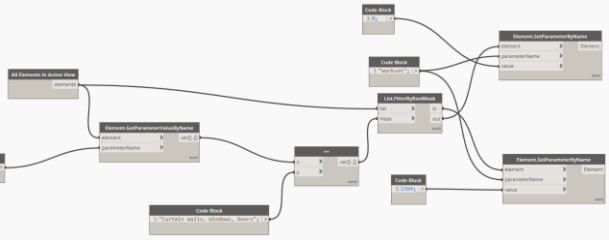
# Dynamo Scripts

Developed dozens of Dynamo scripts for model checking, parameter automation, data extraction, and information analysis across multiple projects. These workflows reduced manual effort, improved team productivity, and supported BIM environment management through automated validation and quality control.

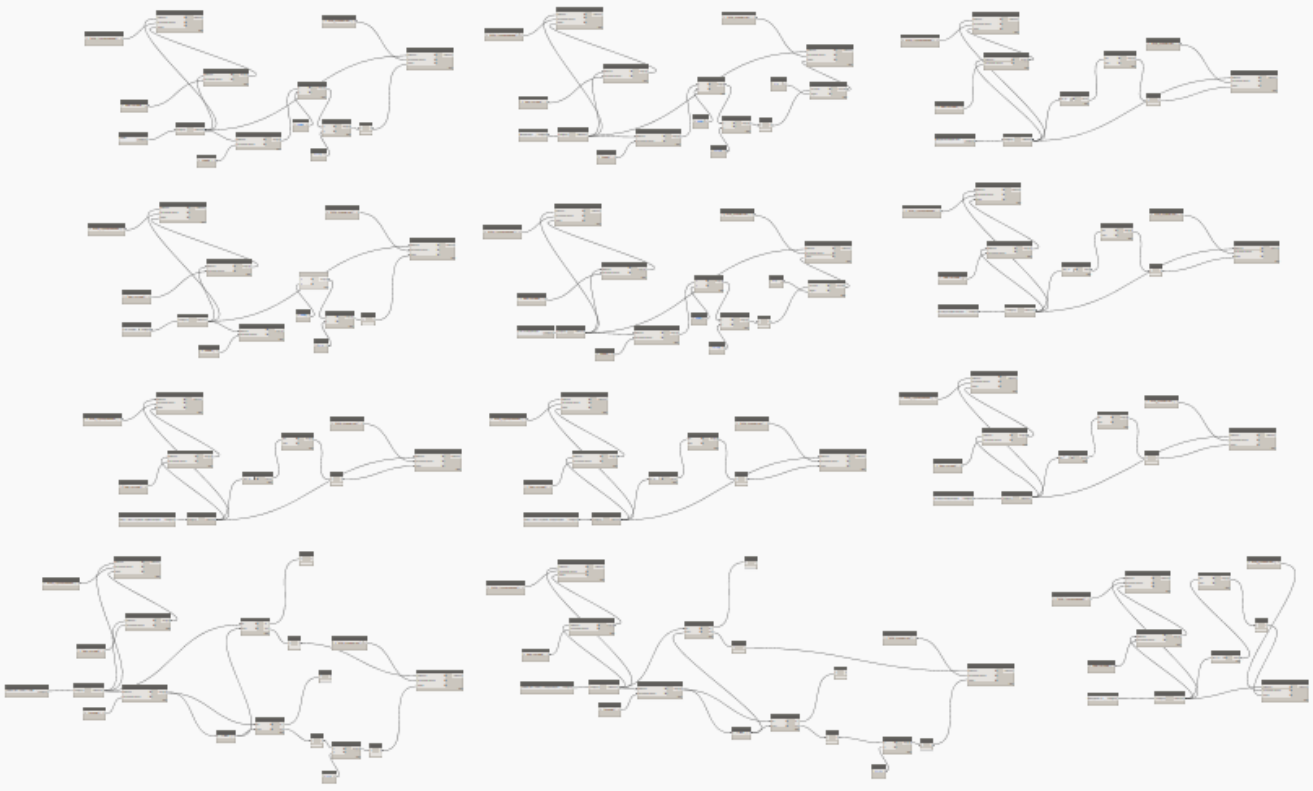
Name Generation Script



Workset Assignment Script

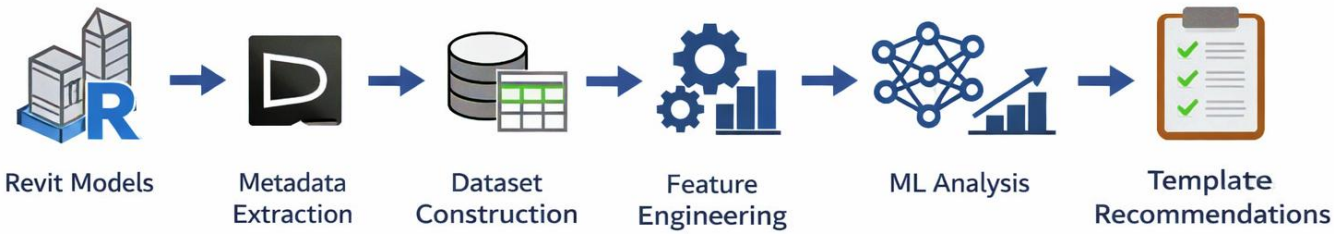


Parameter grouping

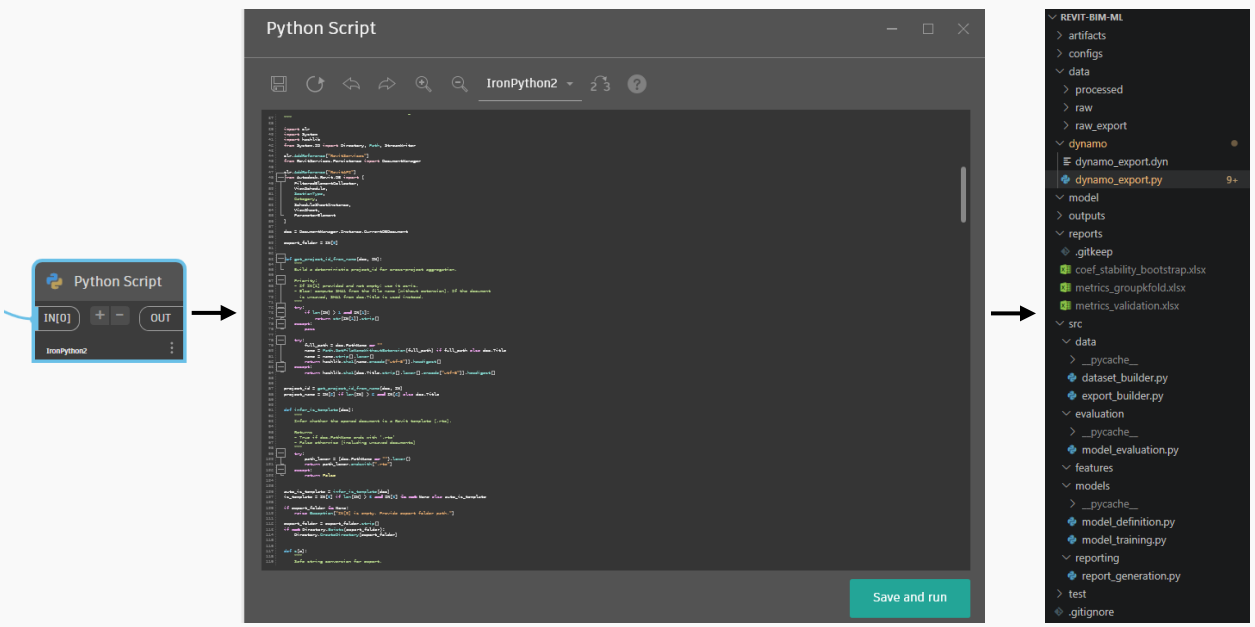


# ML-Based Data Analysis System

My project explores the use of data analytics and machine learning methods to analyze BIM data structures across multiple projects. The main goal was to develop a workflow for extracting structured metadata from Revit schedules and transforming it into a unified analytical dataset. The system evaluates schedule characteristics such as rows, fields, filters, fill rates, and placement on sheets in order to identify reusable patterns. This approach supports BIM template development, standardization, and more consistent project environments.




Data extraction from Revit projects using Dynamo



# ML-Based Data Analysis System

Using metadata extracted from 50 Revit projects, I built a unified analytical dataset containing over 10,000 schedules described by 26 key parameters. Through feature engineering, raw schedule metadata was converted into indicators capturing structural, content, and usage patterns. These features were then used to train a machine learning model on 2,000 schedule signatures, allowing it to estimate how closely each schedule matched a reusable template-like pattern rather than a project-specific configuration. After model tuning and validation of the results, prediction accuracy was improved from 55% to 80%. The analysis showed that more than 10% of Revit schedules may be suitable for inclusion in project templates, supporting BIM template optimization and standardization. Ranked analytical outputs and automated reports reduced schedule analysis time by over 10x for a single model and over 100x across large model sets.




## Revit Metadata

| project_id | project_na | schedule_i | is_te | signa | signature_s | signa | signature_s | fields | has_q | filters | sorts_ | rows_ | place | visibl | hidde | num_ | text_c | formu | fill_ra | fill_ra | formu | fill_ra |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1cf18d3b5 | VP_01_AR_ | 34989 | 0 | CAT: | 95e5ec81 | CAT: | 7e709462 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1cf18d3b5 | VP_01_AR_ | 64929 | 0 | CAT: | 7480d03d | CAT: | e96480f41 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1cf18d3b5 | VP_01_AR_ | 64975 | 0 | CAT: | 82e074da | CAT: | 9484dd65 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1cf18d3b5 | VP_01_AR_ | 64982 | 0 | CAT: | 7480d03d | CAT: | e96480f41 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1cf18d3b5 | VP_01_AR_ | 74601 | 0 | CAT: | dd7c5d3a | CAT: | 93d7af30c | 11 | 0 | 0 | 3 | 302 | 0 | 10 | 1 | 0 | 11 | 0 | 0 | 0,61 | 0 | 0,61 |
| 1cf18d3b5 | VP_01_AR_ | 74900 | 0 | CAT: | bd45c882 | CAT: | e7bc8459 | 10 | 0 | 0 | 2 | 18 | 0 | 7 | 3 | 0 | 10 | 0 | 0 | 0,79 | 0 | 0,79 |

## Processed Training Data

| signature_extra_soft | signature_soft_hash | appe | total_ | rows_co | fields | filte | has_ | sort | qty | name | place | visibl | hidde | num_ | text_c | formu | fill_ra | fill_ra | formul | fill_rate |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CAT:Multi-Category\|Q0\|F10\|Fl2\|S0\|Fr60 | bd9535da1c2e12310dc | 1 | 1 | 2 | 7 | 2 | 0 | 0 | 0 | 1 | 0 | 7 | 0 | 0 | 7 | 0 | 0 | 0,5 | 0 | 0,5 |
| CAT:Multi-Category\|Q0\|F10\|Fl2\|S1\|Fr60 | 1555df18eff4c4645cc7 | 1 | 1 | 2 | 7 | 1 | 0 | 1 | 0 | 1 | 0 | 7 | 0 | 0 | 7 | 0 | 0 | 0,5 | 0 | 0,5 |
| CAT:Multi-Category\|Q0\|F10\|Fl2\|S1\|Fr80 | f543c595145944ab035 | 1 | 1 | 6 | 7 | 1 | 0 | 1 | 0 | 1 | 1 | 6 | 1 | 0 | 7 | 0 | 0 | 0,72 | 0 | 0,722 |
| CAT:Multi-Category\|Q0\|F10\|Fl2\|S3\|Fr80 | 1c59c983d49f0628040 | 1 | 2 | 4 | 9 | 1 | 0 | 2 | 0 | 1 | 1 | 5 | 4 | 0 | 9 | 0 | 0 | 0,7 | 0 | 0,7 |
| CAT:Multi-Category\|Q0\|F10\|Fl5\|S3\|Fr100 | 921d67f38e11c5763db | 1 | 1 | 9 | 8 | 4 | 0 | 2 | 0 | 1 | 1 | 5 | 3 | 0 | 8 | 0 | 0 | 0,82 | 0 | 0,822 |
| CAT:Multi-Category\|Q0\|F2\|Fl0\|S3\|Fr60 | 5a54d670ef698ab5468 | 1 | 1 | 493 | 2 | 0 | 0 | 2 | 0 | 1 | 0 | 2 | 0 | 0 | 2 | 0 | 0 | 0,6 | 0 | 0,597 |

## Processed Data

| | | | | | | | | | | | | | | | | | | | | | | decision |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| sign | signa | total | rows_ | fields_( | filter | has_ | sort | qty_s | fill_rate | place( | visible | hidde | num | text_ | formu | fill_rate_r | fill_rat | formul | appe | tota | appe | in_ | P_to_ | auto_include |
| CAT | 126d5 | 3 | 8 | 179 | 1 | 0 | 1 | 1 | 0,2657 | 0 | 135 | 44 | 6 | 22 | 151 | 0 | 0 | 0,266 | 7 | 7 | 1 | 0 | 0,974 | auto_include |
| CAT | e6eb | 1 | 41 | 131 | 2 | 0 | 4 | 1 | 0,5171 | 0 | 15 | 116 | 8 | 29 | 94 | 0 | 0,21 | 0,671 | 3 | 7 | 0,43 | 0 | 0,938 | auto_include |
| CAT | c8d3f | 6 | 7,5 | 161,5 | 4 | 0 | 1 | 1 | 0,27775 | 0 | 119,5 | 42 | 5 | 21 | 135,5 | 0 | 0 | 0,278 | 3 | 7 | 0,43 | 0 | 0,932 | auto_include |
| CAT | 4520e | 1 | 5 | 124 | 1 | 0 | 1 | 1 | 0,2895 | 0 | 38 | 86 | 9 | 11 | 104 | 0 | 0 | 0,29 | 7 | 7 | 1 | 0 | 0,921 | auto_include |
| CAT | 5627 | 1 | 5013 | 12 | 0 | 0 | 0 | 0 | 0,6574 | 0 | 12 | 0 | 0 | 12 | 0 | 0 | 0,657 | 0 | 1 | 7 | 0,14 | 0 | 0,914 | auto_include |
| CAT | 001ce | 3 | 5 | 123 | 1 | 0 | 1 | 1 | 0,2895 | 0 | 38 | 85 | 9 | 10 | 104 | 0 | 0 | 0,29 | 7 | 7 | 1 | 0 | 0,913 | auto_include |
| CAT | 903a6 | 1 | 9 | 167 | 1 | 0 | 2 | 1 | 0,4444 | 0 | 5 | 162 | 5 | 19 | 143 | 0 | 0,333 | 0,472 | 1 | 7 | 0,14 | 0 | 0,896 | auto_include |
| CAT | d8d8 | 1 | 6 | 108 | 4 | 0 | 4 | 1 | 0,2222 | 0 | 12 | 96 | 6 | 25 | 77 | 0 | 0,333 | 0,2 | 6 | 7 | 0,86 | 0 | 0,89 | auto_include |
| CAT | 6db5 | 4 | 7 | 130 | 4 | 0 | 1 | 1 | 0,2955 | 0,25 | 102 | 28 | 5 | 17 | 108 | 0 | 0 | 0,296 | 3 | 7 | 0,43 | 0 | 0,886 | auto_include |
| CAT | fa7ec | 1 | 40 | 10 | 1 | 0 | 2 | 1 | 0,9857 | 0 | 7 | 3 | 2 | 6 | 2 | 1 | 0,967 | 1 | 2 | 7 | 0,29 | 0 | 0,872 | auto_include |
| CAT | e149 | 2 | 121 | 5 | 0 | 0 | 4 | 1 | 0,88935 | 0 | 4 | 1 | 1 | 3 | 1 | 0,8295 | 0,949 | 0,83 | 2 | 7 | 0,29 | 0 | 0,867 | auto_include |
| CAT | 4d105 | 5 | 16 | 98 | 2 | 0 | 4 | 1 | 0,5357 | 0,2 | 14 | 85 | 8 | 28 | 62 | 0 | 0,438 | 0,575 | 3 | 7 | 0,43 | 0 | 0,864 | auto_include |
| CAT | 4d105 | 3 | 16 | 125 | 6 | 0 | 3 | 1 | 0,6146 | 0 | 12 | 113 | 8 | 28 | 89 | 0 | 0,781 | 0,581 | 2 | 7 | 0,29 | 0 | 0,863 | auto_include |
| CAT | 4d105 | 2 | 15 | 111 | 6 | 0 | 3 | 1 | 0,5667 | 0 | 12 | 99 | 8 | 28 | 75 | 0 | 0,567 | 0,567 | 2 | 7 | 0,29 | 0 | 0,859 | auto_include |
| CAT | fbaa7 | 2 | 1197 | 6 | 0 | 0 | 4 | 0 | 0,97005 | 0 | 6 | 0 | 0 | 6 | 0 | 0 | 0,97 | 0 | 4 | 7 | 0,57 | 0 | 0,85 | auto_include |
| CAT | ffcbe | 1 | 57 | 10 | 1 | 0 | 3 | 1 | 0,8647 | 0 | 7 | 3 | 3 | 5 | 2 | 0,8421 | 0,87 | 0,86 | 1 | 7 | 0,14 | 0 | 0,848 | auto_include |

# THANK YOU FOR YOUR TIME

garkushenkovladimir@gmail.com | LinkedIn: vladimir-garkushenko